ARIEL-SAAS™ AGRO-INDUSTRIAL TELEMETRY SUITE SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Document Code: AREL-SRS-AGRO-2026-V1
Target Portal: energy.arielmc.com (Port 3002)
Client Facility: GreenCycle 22.0 Bigha
Division: AREL & Ariel-SaaS™

1. Executive Summary & Architectural Scope

This SRS establishes the functional, telemetry, automation, and SCADA requirements for deploying the **Ariel-SaaS™ Agro-Industrial Suite** on energy.arielmc.com (Port 3002). The software operates an edge-to-cloud topology uniting (1) Agro-Biological Environmental Telemetry (Milk 4.0°C chilling, Aquaculture optical DO, precision drip NPK, mushroom humidity, poultry NH3) with (2) Sovereign Clean Energy Microgrid SCADA (80 kWp Solar PV, 120 kWh BESS, 25 m³ Biogas, sub-20ms islanding, ISO 14064 carbon ledger).

2. User Roles & Role-Based Access Control (RBAC)

Role Identifier	Role Name	Operational Scope & Permissions
ROLE_SUPER_ADMIN	Ariel Enterprise Admin	Root control across energy.arielmc.com, tenant provisioning, firmware updates, sensor calibration offsets.
ROLE_FARM_OWNER	Executive Farm Owner	Read-write access to executive HUD, financial P&L;, debt servicing reports, ESG audit sheets, security cameras.
ROLE_FARM_GM	Farm General Manager	Full operational command: alarm threshold settings, emergency manual overrides, daily workforce logs.
ROLE_VET_HERD	Veterinarian & Herd Specialist	Dairy bulk milk chilling graphs (strict 4.0°C), cattle/goat vaccination schedules, poultry FCR logs.
ROLE_AQUA_SPEC	Aquaculture Specialist	Pond optical DO & pH telemetry, automated aerator start thresholds, feed conversion tracking.
ROLE_MICROGRID_ENG	Resident Electrical Engineer	Solar string telemetry, 94.2% PR tracking, BESS cell-level voltage balancing, biogas pressure, AMF generator.
ROLE_AUDITOR_PFI	Bank / ESG Compliance Officer	Read-only access to power generation logs, ISO 14064 carbon certificates, milk off-take escrow receipts.

3. Functional Modules & Automated Field Closed-Loop Triggers

Module Identifier	Key Functional Requirements & Automation Relays	Target Thresholds & Siren Rules
Mod 1: Dairy Chilling	Polls dual RTD PT100 probes every 30s. Logs daily milk liters per cow with RFID tag integration.	Strict **4.0°C** target. SMS siren triggers if temp >4.8°C for >10 mins.
Mod 2: Aquaculture DO	Optical DO (0-20 mg/L) & pH polling every 60s. Auto starts 2 HP paddlewheel aerator relay via wireless latch.	Starts aerator when DO **<3.2 mg/L**; stops at **>5.5 mg/L**. Siren if DO <2.5 mg/L.
Mod 3: Precision Drip	6-in-1 Soil NPK & moisture sensors. Automated 2-inch DC latching solenoid valves for pulsed fertigation.	Starts drip when soil VWC **<22%**. Locks out if rain sensor registers >5mm precipitation.
Mod 4: Indoor Climate	Mushroom NDIR CO2 & RH sensors; poultry electrochemical NH3 sensor. Automatic ultrasonic fogger control.	Maintains **85–90% RH** in darkrooms. Auto fan triggers if NH3 >20 ppm in poultry shed.
Mod 5: Clean SCADA	Sub-second Modbus polling of Huawei inverter, BESS PCS, and Class-A Pyranometer. SREDA net-metering sync.	Tracks **94.2% PR**. Detects islanding in **<20ms**. Auto-starts biogas gen if BESS <25%.
Mod 6: Sirens & Alerts	Multi-channel dispatch via WebSockets, SMS (SSL Wireless), WhatsApp Webhook, and audible browser sirens.	Sub-second alert dispatch. Escalates to Farm Owner if unacknowledged >15 mins.
Mod 7: ESG Carbon Desk	Continuous avoided CO2e ledger (Solar kWh x 0.62 kg + Biogas CH4 x 28x GWP = ~120 MT CO2e avoided/yr).	One-click signed tamper-proof PDF audit sheets for **ISO 14064** compliance.

4. Technical Stack & Edge-to-Cloud Topology

- **Edge Gateway:** DIN-rail Linux Gateway with SX1302 LoRaWAN concentrator (ChirpStack Bridge) + RS485 Modbus RTU.
- **Cloud Ingestion:** EMQX MQTT Broker with TLS 1.3 encryption on energy.arielmc.com.
- **Storage:** TimescaleDB (time-series sensor packets, PR, SoC) + PostgreSQL (RBAC, financial ledgers, audit trails).
- **Application Layer:** Next.js 14 App Router, TypeScript, Tailwind CSS, WebSockets streaming.
- **Offline Autonomy:** Edge gateway buffers 30 days of data locally during rural cellular outages with auto-replay.

Certified for Software Development & Architecture:
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Ariel-SaaS™ Engineering Division | energy.arielmc.com (Port 3002)